

EXPERIMENT 8

AIM: Programs development using creation of procedures, passing parameters IN and OUT of PROCEDURES

SOLUTION:

Procedure in PL/SQL:

A **procedure** is a named PL/SQL block that is stored in the database and can be executed again and again.

- ✓ It is mainly used to **perform an action** (insert, update, display, calculate, etc.)

Syntax of Creating a Procedure

```
CREATE OR REPLACE PROCEDURE procedure_name (  
    parameter_name mode datatype,  
    parameter_name mode datatype  
)  
IS  
BEGIN  
    statements;  
END;  
/
```

Modes:

- ✓ IN → Input only (default)
- ✓ OUT → Output only
- ✓ IN OUT → Input + Output

IN Parameter:

Example: Procedure to find square of a number

```
CREATE OR REPLACE PROCEDURE find_square  
(p_num IN NUMBER)  
IS  
BEGIN  
    DBMS_OUTPUT.PUT_LINE('Square is: ' || (p_num * p_num));  
END;  
/
```

Execute Procedure:

```
SET SERVEROUTPUT ON;
```

```
EXEC find_square(5);
```

OUT Parameter

Example: Procedure to return result

```
CREATE OR REPLACE PROCEDURE find_cube  
(p_num IN NUMBER, p_result OUT NUMBER)  
IS  
BEGIN  
    p_result := p_num * p_num * p_num;  
END;  
/
```

Execute Procedure:

```
DECLARE  
    ans NUMBER;  
BEGIN  
    find_cube(3, ans);  
    DBMS_OUTPUT.PUT_LINE('Cube is: ' || ans);  
END;  
/
```

IN OUT Parameter:

Example: Update value using IN OUT

```
CREATE OR REPLACE PROCEDURE double_value  
(p_num IN OUT NUMBER)  
IS  
BEGIN  
    p_num := p_num * 2;  
END;  
/
```

Execute the Procedure:

```
DECLARE  
    x NUMBER := 10;  
BEGIN  
    double_value(x);  
    DBMS_OUTPUT.PUT_LINE('Result is: ' || x);
```

END;

/

Program :

Create table

```
CREATE TABLE student_proc (  
    rollno NUMBER PRIMARY KEY,  
    name VARCHAR2(20),  
    marks NUMBER  
);
```

Insert data:

```
INSERT INTO student_proc VALUES (1, 'Ravi', 75);  
INSERT INTO student_proc VALUES (2, 'Anu', 45);
```

COMMIT;

Create Procedure (IN + OUT):

```
CREATE OR REPLACE PROCEDURE student_result  
(  
    p_roll IN NUMBER,  
    p_name OUT VARCHAR2,  
    p_result OUT VARCHAR2  
)  
IS  
    v_marks NUMBER;  
BEGIN  
    SELECT name, marks INTO p_name, v_marks  
    FROM student_proc  
    WHERE rollno = p_roll;  
    IF v_marks >= 50 THEN  
        p_result := 'PASS';  
    ELSE  
        p_result := 'FAIL';  
    END IF;  
EXCEPTION  
    WHEN NO_DATA_FOUND THEN
```

```
p_name := 'NOT FOUND';  
p_result := 'INVALID ROLL NUMBER';  
END;  
/
```

Execute Procedure:

```
SET SERVEROUTPUT ON;  
  
DECLARE  
    v_name student_proc.name%TYPE;  
    v_res VARCHAR2(20);  
  
BEGIN  
    student_result(1, v_name, v_res);  
    DBMS_OUTPUT.PUT_LINE('Name: ' || v_name);  
    DBMS_OUTPUT.PUT_LINE('Result: ' || v_res);  
END;  
/
```

Output:

Name: Ravi

Result: PASS

PL/SQL procedure successfully completed.